A REMARK ON LOCAL LOGARITHMIC EMBEDDINGS FOR THE NEVANLINNA CLASS OF DIRICHLET SERIES

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Abstract. Allu and Mondal asked whether the local integrals of $\log(1+|f|)$ over unit vertical segments in the right half-plane are bounded by a constant times the Nevanlinna gauge $\|f\|_0$, for Dirichlet series f in the Nevanlinna class $\mathscr{N}_u$. We show that the answer is no, already for exponentials of Dirichlet polynomials, which are bounded in the half-plane: along the family $\exp(\sum_{n=2}^{N} n^{-s})$ the ratio of the two sides is at least of order $\sqrt{N}/\log N$. After rescaling the exponent, the local integrals are unbounded on a set of bounded Dirichlet series with bounded gauge, so the local integral is not bounded by any function of $\|f\|_0$.

1. Introduction

We use the notation of Allu and Mondal [1]. For $\theta \in \mathbb{R}$ let $\mathbb{C}_\theta = \{s : \operatorname{Re} s > \theta\}$, let $f(s) = \sum_{n\geq 1} a_n n^{-s}$ be a Dirichlet series whose abscissa of uniform convergence satisfies $\sigma_u(f) \leq 0$, and let $f_\sigma(s) = f(s+\sigma)$. Allu and Mondal consider the class $\mathscr{N}_u$, introduced by Brevig and Perfekt [2] (see also Guo, Ni and Zhou [4]), of such series with

$$\|f\|_0 = \sup_{\sigma>0} \|f_\sigma\|_0 = \lim_{\sigma\to 0^+} \|f_\sigma\|_0 < \infty, \tag{1}$$

where

$$\|f_\sigma\|_0 = \lim_{T\to\infty} \frac{1}{2T} \int_{-T}^{T} \log\bigl(1+|f(\sigma+it)|\bigr)\,\mathrm{d}t;$$

see [1, (1.4), (1.6) and (2.1)]. The function $\sigma \mapsto \|f_\sigma\|_0$ is nonincreasing [4], and by [1, (2.2)] the gauge is also $\|f_\sigma\|_0 = \int_{\mathbb{T}^\infty} \log(1+|\mathcal{B}f_\sigma|)\,\mathrm{d}m_\infty$, where $\mathcal{B}$ is the Bohr lift and m_∞ is the Haar measure of $\mathbb{T}^\infty$. We write χ for a character of $\mathbb{T}^\infty$, that is, a completely multiplicative function on the positive integers with $|\chi(n)| = 1$, so that $\mathcal{B}f(\chi) = \sum_n a_n\chi(n)$.

Motivated by local Carleson embeddings for Hardy spaces of Dirichlet series, they posed the following question [1, Problem 4.13].

Problem 1 (Allu and Mondal)**.** *Is there a constant $C > 0$ such that every $f \in \mathscr{N}_u$ with $\sigma_u(f) \leq 0$ satisfies*

$$\sup_{\sigma>0} \sup_{\tau\in\mathbb{R}} \int_{\tau}^{\tau+1} \log\bigl(1+|f(\sigma+it)|\bigr)\,\mathrm{d}t \leq C\|f\|_0\,? \tag{2}$$

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The purpose of the present note is to show that the answer is negative.

Theorem 1. *For $N \geq 3$ let $f_N(s) = \exp(\sum_{n=2}^N n^{-s})$. Then f_N is a Dirichlet series with $\sigma_u(f_N) = -\infty$, it is bounded in $\overline{\mathbb{C}_0}$, and*

$$\|f_N\|_0 \leq \log 2 + \frac{\sqrt{N-1}}{2\sqrt{2}}, \qquad \int_0^1 \log(1+|f_N(1/\log N + it)|)\,\mathrm{d}t \geq \frac{\pi(N-1)}{6e\log N}.$$

In particular, (2) fails for every C, also for the bounded Dirichlet series in $\mathscr{N}_u$.

The mechanism is simple. On a short vertical segment near the real axis, all the terms n^{-s} with $n \leq N$ point in nearly the same direction, so $\operatorname{Re}\sum n^{-s}$ is of order N on a segment of length of order $1/\log N$. The gauge, on the other hand, is a mean over the whole line, that is, over $\mathbb{T}^\infty$, where the characters $\chi(n)$ are orthogonal and their sum has size $\sqrt{N}$.

2. Proof of Theorem 1

Put $g_N(s) = \sum_{n=2}^N n^{-s}$. Expanding the exponential and collecting products $n_1 \cdots n_k$, we find that $f_N = e^{g_N}$ is a Dirichlet series $\sum_m a_m m^{-s}$ with $a_m \geq 0$. For every real θ, $\sum_m a_m m^{-\theta} = \exp(g_N(\theta)) < \infty$, so the series converges absolutely and uniformly in $\overline{\mathbb{C}_\theta}$. Hence $\sigma_u(f_N) = -\infty$, and $|f_N(s)| \leq e^{N-1}$ in $\overline{\mathbb{C}_0}$. In particular $f_N \in \mathscr{N}_u$.

The gauge. Fix $\sigma > 0$. The Bohr lift of $(f_N)_\sigma$ is

$$\mathcal{B}(f_N)_\sigma(\chi) = \exp\Big(\sum_{n=2}^N n^{-\sigma}\chi(n)\Big), \qquad X_\sigma(\chi) = \operatorname{Re}\sum_{n=2}^N n^{-\sigma}\chi(n),$$

so that $|\mathcal{B}(f_N)_\sigma| = e^{X_\sigma}$. By [1, (2.2)],

$$\|(f_N)_\sigma\|_0 = \int_{\mathbb{T}^\infty} \log(1+e^{X_\sigma})\,\mathrm{d}m_\infty \leq \log 2 + \mathbb{E}(X_\sigma)_+,$$

where $\mathbb{E}$ denotes integration against m_∞ and we used $\log(1+e^x) \leq \log 2 + \max(x,0)$. For f_N this identity is also a direct consequence of Kronecker's theorem: X_σ depends only on the coordinates $\chi(p)$, $p \leq N$, and is continuous in them, and since the numbers $\log p$ are linearly independent over $\mathbb{Q}$, the curve $t \mapsto (p^{-it})_{p\leq N}$ is equidistributed in the finite torus.

By unique factorization the functions $\chi \mapsto \chi(n)$, $2 \leq n \leq N$, are distinct nonconstant characters of $\mathbb{T}^\infty$, and the products $\chi(n)\chi(n') = \chi(nn')$ are nonconstant. Writing $X_\sigma = (G + \overline{G})/2$ with $G = \sum_n n^{-\sigma}\chi(n)$, we therefore have $\mathbb{E}G = \mathbb{E}G^2 = 0$ and $\mathbb{E}|G|^2 = \sum_n n^{-2\sigma}$, so

$$\mathbb{E}X_\sigma = 0, \qquad \mathbb{E}X_\sigma^2 = \frac{1}{2}\sum_{n=2}^N n^{-2\sigma} \leq \frac{N-1}{2}.$$

Since $\mathbb{E}X_\sigma = 0$, the Cauchy–Schwarz inequality gives $\mathbb{E}(X_\sigma)_+ = \mathbb{E}|X_\sigma|/2 \leq (\mathbb{E}X_\sigma^2)^{1/2}/2 \leq \sqrt{N-1}/(2\sqrt{2})$. This holds for every $\sigma > 0$, and (1) gives the bound for $\|f_N\|_0$.

The local integral. Let $\sigma = 1/\log N$ and $0 \le t \le \pi/(3\log N)$. For $2 \le n \le N$ we have $n^{-\sigma} \ge N^{-1/\log N} = e^{-1}$ and $0 \le t\log n \le \pi/3$, so

$$\operatorname{Re} n^{-\sigma-it} = n^{-\sigma}\cos(t\log n) \ge \frac{1}{2e}.$$

Hence $\log(1+|f_N(\sigma+it)|) \ge \operatorname{Re} g_N(\sigma+it) \ge (N-1)/(2e)$ on an interval of length $\pi/(3\log N) \le 1$, and the integral over $[0,1]$ is at least $\pi(N-1)/(6e\log N)$.

Finally, the lower bound for the local integral divided by the upper bound for $\|f_N\|_0$ tends to infinity as $N \to \infty$. So no constant C satisfies (2) for all f_N. □

The bound for the gauge holds for every $\sigma > 0$, and by continuity also at $\sigma = 0$. It therefore applies to each of the equivalent definitions of the gauge in [1], and also with $\log^+$ in place of $\log(1+\cdot)$.

3. No bound by a function of the gauge

Rescaling the exponent shows that the local integral is not controlled by the gauge in any way.

Corollary 1. *For $N \ge 3$ let $F_N(s) = \exp\bigl((N-1)^{-1/2}\sum_{n=2}^{N} n^{-s}\bigr)$. Then $F_N \in \mathscr{N}_u$ is bounded in $\overline{\mathbb{C}_0}$, and*

$$\|F_N\|_0 \le \log 2 + \frac{1}{2\sqrt{2}}, \qquad \int_0^1 \log(1+|F_N(1/\log N + it)|)\,\mathrm{d}t \ge \frac{\pi\sqrt{N-1}}{6e\log N}.$$

In particular, there is no function ψ such that the left-hand side of (2) *is at most $\psi(\|f\|_0)$ for every $f \in \mathscr{N}_u$.*

Proof. In the proof of Theorem 1, replace X_σ by $X_\sigma/\sqrt{N-1}$. Then the expectation of its positive part is at most $1/(2\sqrt{2})$, which gives the first bound. For the second, use $\log(1+e^x) \ge x$: on the interval used in the proof of Theorem 1, the real part of the exponent of F_N is at least $\sqrt{N-1}/(2e)$. □

4. Concluding remarks

Functions of one prime. Every function of 2^{-s} alone satisfies (2) with $C = 2\pi/\log 2$, since $t \mapsto \log(1+|f(\sigma+it)|)$ is then periodic with period $2\pi/\log 2 > 1$. So a counterexample must involve an unbounded number of frequencies, as ours do.

Relation to Hardy spaces. The local embedding problem for $\mathscr{H}^p$, which Chen, Fang, Guo, Hou, Shao and Zhou [3] have recently announced for every $p > 2$, concerns the line $\operatorname{Re} s = 1/2$ and the L^p norm. In (2) the segments approach the line $\operatorname{Re} s = 0$, where a Dirichlet polynomial with N terms can be of size N on a whole unit segment, while its root mean square over the line is only $\sqrt{N}$; the logarithm does not compensate for this.

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References

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